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	POSTDOCTORAL FELLOWSHIP Spatial orientation is achieved by integrating sensory information from different modalities in order to estimate both body orientation and self-motion. Multiple sources of relevant sensory information are often available, so the question becomes, how should the nervous system combine them? If the goal is to achieve the single most probable combined estimate, then the rule is simple; each information source should be weighted based on its relative reliability. This kind of statistically optimal combination rule is known as Bayesian estimation, and it has been shown to accurately characterize perception in a variety of situations. We aim to develop a comprehensive Bayesian model for spatial orientation perception. This probabilistic model will take visual and vestibular signals as input, and it will generate combined estimates of orientation and linear and angular		

Task Description:	self-motion as output. The model will make specific, testable predictions about how visual and vestibular information should be combined. It will also make predictions about the time course of perceptual estimates in response to time-varying visual and vestibular inputs. In order to develop the best possible model, we have conducted a thorough review of existing spatial orientation models. In particular, we have focused on statistically optimal models of spatial orientation that are dynamic, meaning that the model input and output are continuous and vary over time. The two approaches that appear most promising are Kalman and Particle filtering techniques. A Kalman filter is a control system architecture that compares predicted and observed output at each time step and uses the result to drive the dynamics of the system. It is statistically optimal because the gain applied to the feedback signal is influenced by the reliability of the input signals. A Particle filter is a probabilistic modeling technique that relies on simulating the full probability distributions of all inputs and outputs at each time step. In order to evaluate these two alternative approaches we have simulated them. The results of these simulations are guiding development of a revised dynamic Bayesian model. We are also conducting a number of psychophysical experiments to measure discrimination performance for a variety of spatial orientation stimuli. These experiments use a motion simulator and measure perceptual estimates of orientation and linear and angular self-motion in visual-only, vestibular-only, and combined visual-vestibular conditions. These reliability measures are important parameters of statistically optimal models. We use a two-alternative-forced-choice procedure because this method minimizes potential sources of response noise and bias. Also, because all data is collected using a common method and apparatus, it is possible to make interesting comparisons across conditions. Recent experiments have investigated perception of heading, which is the direction of self-motion. Heading perception is fundamental to effective navigation and vehicle guidance, so it is important to understand the factors that influence the precision of visual and vestibular heading estimates. We measured discrimination thresholds for heading azimuth and elevation in visual-only and vestibular-only conditions with observers oriented upright and side-down relative to gravity. Visual thresholds were significantly lower than vestibular, and upright thresholds were generally lower than side-down. Both visual and vestibular results revealed that observers are better at discriminating head-centric azimuth than elevation, regardless of body orientation. In other words, sensitivity to heading depends more upon the direction of self-motion relative to the head, than relative to gravity. Another ongoing experiment is investigating how the threshold for vestibular detection of linear self-motion depends on the direction of translation in head coordinates. We plan to conduct a similar series of experiments to investigate how vestibular detection of angular self-motion depends on the axis of rotation. Linear and angular self-motion thresholds are measurements of the reliabilities of vestibular sensory estimates which are necessary parameters of statistically optimal models. We are also conducting an experiment to investigate the interaction of vestibular rotation and translation signals for the perception of self-motion on a curved path. Results of this experiment will be compared with predictions of the model, once it is developed. In future we plan to conduct a similar experiment to investigate the perception of tilt versus translation. The research described above will contribute to a better understanding of spatial orientation in general. The model may be used to predict and investigate situations where spatial disorientation is likely to occur. The results of the experiments may be applied to the development of countermeasures including better cockpit display technology, improved motion simulations, novel pilot training techniques, and crew screening procedures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Spatial orientation allows astronauts to pilot the space shuttle and navigate their way around the inside of the international space station. It also allows people here on earth to get where they are going and avoid collisions while driving a car or riding a bike. Perception of spatial orientation is essential for mobile organisms to navigate effectively within the environment. Without perception of spatial orientation we are literally lost. By successfully modeling perception of spatial orientation we can predict when spatial disorientation is most likely to occur. This will help astronauts avoid potentially disorienting and hazardous situations in space. It could also help avoid spatial disorientation here on earth. For example, it could lead to the development of technology that helps drivers avoid disorientation and accidents on the road. The psychophysical measurements we are making also contribute to a better general understanding of spatial orientation perception. The methodologies we have developed for these experiments could be used to assess performance of spatial orientation tasks by astronauts. They could also be used to assess the performance of aircraft pilots or patients with visual or vestibular deficits.
Task Progress:	Modeling progress: We have reviewed most existing models of spatial orientation perception and simulated the two of them that have the most in common with the statistically optimal model we aim to develop, namely the Kalman filter model proposed by Borah et al. (1988) and the Particle filter model proposed by Laurens & Droulez (2007). The results of these simulations are guiding development of a novel, dynamic and statistically optimal model of spatial orientation perception. Psychophysics progress: We have completed a series of experiments investigating human ability to perceive the direction of self-motion based on visual and vestibular cues. The visual cue to heading is the location of the focus of expansion in the optic flow field. The vestibular cue to heading is the direction of inertial acceleration signaled by the otoliths. Prior research has focused on visual discrimination of heading azimuth (heading in the horizontal plane). There have been few studies of visual discrimination of elevation, and fewer comparable studies of non-visual heading discrimination. To investigate human ability to estimate heading under more general conditions, we measured heading discrimination thresholds for azimuth and elevation in visual-only and vestibular-only conditions with observers oriented upright and side-down relative to gravity. Experiments were conducted in the Human Motion Lab at Washington University, a motion simulator consisting of a 6DOF Moog motion base and 90 deg X 90 deg stereo projection screen. Subjects were seated on the motion platform in a padded racing seat, and held in place with a 5-point harness. The head was secured to a cushioned head mount by a form fitted plastic mesh mask. Noise was played through noise-cancellation headphones and responses were collected using a custom made button box. Subjects were asked to discriminate heading azimuth or elevation relative to straight-ahead in a two-interval-forced-choice task. Visual thresholds were significantly lower than

	vestibular, and upright thresholds were generally lower than side-down. Both visual and vestibular results revealed that observers are better at discriminating head-centric azimuth than elevation, regardless of body orientation. In other words, sensitivity to heading depends more upon the direction of self-motion relative to the head, than relative to gravity. Heading perception is fundamental to effective navigation and vehicle guidance, so it is important to understand how the precision of visual and vestibular heading estimates depend on the direction of self-motion in head-coordinates and world coordinates. Furthermore, it is useful to characterize the reliabilities of visual and vestibular heading estimates because these are necessary parameters of statistically optimal models like the one we aim to develop.
Bibliography Type:	Description: (Last Updated: 02/11/2010)
Abstracts for Journals and Proceedings	Butler JS, MacNeilage PR, Campos JL, Bülthoff HH. "Optic flow velocity profiles influence heading and speed discrimination." 30th European Conference of Visual Perception, Arezzo, Italy, August 27-31, 2007. Perception 2007;36 Suppl:86. , Aug-2007
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